IAE PARIS — SORBONNE BUSINESS SCHOOL · IDGP ACADEMY

Digital Tool Appraisal

The instrument for the afternoon's group activity. One tool, about twenty minutes, filled in by the people who will live with the decision.

17 October 2026 · 14:00–17:00

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Companion to the Digital Tools & Benchmarking reading notes

This printed version is for working in the room. The same instrument is fillable and exportable online, which is the version to use when the appraisal has to leave the session.

Contents

Does the tool change what the organisation owns or what it is able to do; where it contains AI, what the system was told to want, to know and to see, and at which link the chain from a working model to a result breaks; and which capability gaps to close first.

ONE **What you need in order to answer** 4

TWO **The instrument** 6

THREE **Reading the result** 16

FOUR **Sources** 17

> **There is no score.** The instrument restates your own answers and locates the gap. A composite figure would hide the only thing worth knowing, which is where the chain stops.

Three questions decide whether a digital investment returns anything. Does it change what the organisation *owns*, or what it is *able to do*? At which link does the chain from a working tool to an organisational result break — or go unmeasured? And, where the tool contains AI, what has the system been told to want, to know and to see?

The instrument asks all three in the room, in about twenty minutes for one tool, and then turns outward in two follow-up sections: what a reference organisation does that you do not, and which capability gaps to close first over six months. It is designed to be filled in by the people who will live with the decision rather than by the people selling it, and **"not documented" and "nobody knows" are its most useful answers.**

ONE

What you need in order to answer

Everything below is developed in the afternoon's reading notes. This page carries only the three ideas the fields depend on.

A tool enters the bracket; it does not touch the coefficient

$$V = C \times (A + P)$$

assets and processes are additive and purchasable · capability is a coefficient

A platform is almost always an **A** decision — the organisation owns something new. Sometimes it is also a **P** decision — the way work flows changes. It is very rarely a **C** decision, which would mean the organisation ends up better at *improving* its processes than it was before. The error that destroys business cases is paying for A and booking the benefit as though C had moved.

Six conditions between a working tool and a result

0
Evidence base
THE TOOL

1
Technical capability in operation
THE TOOL

2
Workflow integration
CAPABILITY 1

3
Translation into action
CAPABILITY 2

4
Organisational outcome
THE RESULT

5
Net value
THE RESULT

Each is necessary and none is sufficient: a break at any link stops the effect, however good the links before it. The two in the middle are not properties of the tool at all — they are capabilities of the organisation that bought it, and they are where a systematic review of real implementations found value most often lost. The chain is drawn from that review (Lima, manuscript in preparation); its evidence base is hospital implementations and it is used here as a structure for appraisal, not as a citable result.

Where the tool contains AI: three things nobody wrote down

Pillar	The question	What it looks like when it is missing
Teleology	What has the system been told to want — and when its objectives conflict, which one wins?	It optimises one metric brilliantly and nobody can say what it traded away
Epistemology	What is it allowed to count as knowing — and does it know when it does not know?	It returns an answer with high confidence in conditions that no longer resemble its training data
Ontology	What is it able to see — and which variable that matters does not exist for it?	A supplier represented by price and lead time alone, in a decision that turned on geopolitics

After Schrage and Kiron. An assumption nobody wrote down is one nobody can audit or improve — which is why the audit in section 4 asks not whether the system is good, but whether the assumption was *articulated*.

TWO

The instrument

Appraise one tool at a time. Answer from what the organisation can actually evidence, not from what the business case promised. **"Not documented" and "nobody knows" are the most useful answers here** — they are precisely what the instrument is looking for.

Digital Tool Appraisal

In the room: sections 1 to 5, about twenty minutes for one tool. Sections 6 and 7 are the follow-up assignment.

1 · The tool

Tool or platform

What it is called

The workflow it automates

Be specific. "Improves collaboration" is not a workflow.

One line

Where it stands

- [ ] Under evaluation — not yet bought
- [ ] Bought, being implemented
- [ ] In production under a year
- [ ] In production more than a year

Does it include AI or machine learning?

Anything that learns from data, scores, ranks, forecasts, classifies or generates. If a vendor calls it AI and you are not sure what it does, answer yes — that uncertainty is itself a finding.

- [ ] Yes
- [ ] Unsure — the vendor says so; we cannot say what it does
- [ ] No

2 · VCAP triage — what does it actually change?

Which term does it move?

Training people to operate the new system is part of delivering the asset. C means the organisation ends up better at improving its processes than it was before.

- [ ] A — we own something we did not own before
- [ ] A and P — the way work flows changes too
- [ ] A, P and C — we will also be better at improving afterwards
- [ ] Honestly, none of them yet

If you claimed C: what will the organisation be able to do afterwards that it cannot do now — and that does not depend on this system?

Leave blank if you did not claim C. A blank here is a legitimate and common answer.

Does it limit work in progress?

Work in progress is everything started and not yet finished. With the same capacity, more in flight means each item waits longer — Little's Law from the morning. A tool that makes work easier to *start* without capping what is in progress lengthens lead time by construction.

- [ ] Yes — there is an explicit cap, and it is enforced
- [ ] There is a cap that nobody enforces
- [ ] No — it only makes starting work easier

Where will each fact live afterwards?

Every type of fact — a schedule date, a risk, a cost figure — needs exactly one authoritative home, with everything else reading from it. The moment the same fact lives in two systems, somebody reconciles them forever and the dashboards become negotiations.

- [ ] It replaces existing sources — one authoritative home afterwards
- [ ] It adds another place where the same fact lives
- [ ] It consumes from an existing authoritative source

Adoption — expected against actual

Of the people meant to do their work *inside* it — not just log in — what share actually do? Write "too early" if it is not live. A tool used by 40% of its intended users does not deliver 40% of a single-source benefit; it delivers close to none, because the other 60% still has to be reconciled.

e.g. expected 100%, actual about 40%

How long before a competitor with the same budget has the same thing?

This is the test for whether the tool buys advantage or only parity. Anything purchasable is available to a rival at roughly the same price.

- ☐ Months — they would buy the same product
- ☐ A year or two
- ☐ Longer, because of what we would also have to build

3 · Capability maturity — the six conditions

The chain from a working tool to an organisational result, after Lima (manuscript in preparation). It applies to any tool; where the tool contains AI, conditions 0 and 1 bite harder. Each condition is necessary and none is sufficient, so the useful answer is the *first* one that is not a clear yes. **Not measured** is a legitimate and common answer — and a more tractable finding than a failure.

Read across, the six answers *are* a maturity statement, and a more defensible one than a stage number: they say which conditions this organisation can currently satisfy and demonstrate, which it satisfies without evidence, and which it has never observed. There is no composite score here on purpose. A single figure would hide the only thing worth knowing, which is *where* the chain stops.

0 · Evidence base — has this kind of application produced a measured result in a real organisation?

"Kind of application" means the job: demand forecasting, scheduling, document generation, risk scoring, case prioritisation. Not a model, not a simulation, not a pilot that reported accuracy — an implementation somebody measured afterwards, ideally in an organisation like yours.

- ☐ Yes, in a comparable organisation
- ☐ Partly — results exist but diverge
- ☐ No, only modelling or vendor material
- ☐ Not measured, or nobody has looked

1 · Technical capability in operation — does it do here what the demo did, on *our* data, the way we actually run it?

For any tool: has anyone checked, after go-live, that it performs as specified on our volumes and our exceptions? For an AI tool, more sharply: vendor accuracy was measured on the vendor's data with the vendor's settings — has it been checked on ours, with the cut-off we actually use, counting only outputs that tell people something they did not already know?

- ☐ Yes, verified locally after deployment
- ☐ Partly — verified once, not monitored
- ☐ No, or performance has drifted

- [ ] Not measured

2 · Workflow integration — does the output reach whoever decides, while the decision is still open?

The first of the two organisational conditions. Name the person and the step, not the dashboard.

- [ ] Yes — named recipient, inside the decision window
- [ ] Partly — it arrives, but late or to the wrong role
- [ ] No — it is available rather than delivered
- [ ] Not measured — we do not know if it is used

Who receives the output, at which step, and is it in fact used?

Two thirds of staff rarely or never opening the tool is the commonest observed failure, and it is invisible unless somebody asks.

Role · step in the process · evidence of use

3 · Translation into action — is there an owner able to act, with the resources the action requires?

The second organisational condition. If acting on the output needs three approvals, a budget line and a resource nobody has, an earlier warning changes nothing — the constraint was never information.

- [ ] Yes — named owner, resourced, low barrier
- [ ] Partly — owner named, resources contested
- [ ] No — identification without capacity to act
- [ ] Not measured — we do not know the follow-through rate

The action, its owner, and what it depends on

Write the sentence the business case should contain: when the output says X, *who* does *what*, using *what*.

4 · Organisational outcome — which indicator should move, measured how?

One named indicator, one data source, one baseline — and not a satisfaction survey. Where perception and objective measurement were collected together, they diverged.

- [ ] Yes — named indicator, source and baseline
- [ ] Partly — indicator named, no baseline
- [ ] No — benefit stated in general terms
- [ ] Measured only by what users report

5 · Net value — what does the gain cost, and who bears it?

Who is de-prioritised, which work moves to other people or other hours, and what checking the output takes. The hour saved in the back office that reappears as an hour of overtime in the front office is not a gain; it is a relocation.

- ☐ Yes — the displaced cost is identified and accepted
- ☐ Partly — suspected but not quantified
- ☐ No — only the gain has been counted
- ☐ Not measured

How good is the evidence you just used to answer?

Three things make a result trustworthy: a comparison group measured at the same time, an objective measure rather than what people said they felt, and a source that is not selling the tool. Tick the weakest point in what you relied on. Vendor material usually fails at least two of the three.

- ☐ None — a comparison group at the same time, an objective measure, an independent source
- ☐ No comparison group — only before-and-after, so anything else that changed gets credited to the tool
- ☐ Measured by what users reported, not by an objective figure
- ☐ The figures come from the vendor
- ☐ More than one of these

> **Section 4 applies only where the tool contains AI or machine learning.** If the last question of section 1 is answered no, go straight to section 5. In the online version the section appears and disappears on its own.

4 · Philosophical debt audit — AI only

Three pillars, after Schrage and Kiron. The question in each case is not whether the system is good, but whether the organisation has *articulated* the assumption — because an assumption nobody wrote down is one nobody can audit or improve.

Teleology — what it is told to want

The hierarchy of purposes. Most critical, most often skipped.

What is it optimising for, precisely?

The number the system actually tries to maximise or minimise — cost per order, forecast error, time to resolution — not the business goal it was sold against. If nobody can say, write "unknown". That is the finding, not a gap in your answer.

The metric it actually maximises or minimises

Is the hierarchy of objectives documented?

When cost, service level, risk and sustainability conflict, is it written down which wins?

- [ ] Yes — written, owned and reviewed
- [ ] Written once, never revisited
- [ ] Not documented — it is implicit in the configuration
- [ ] Nobody in this organisation could answer it

Epistemology — what it is allowed to count as knowing

How it acquires knowledge, validates it, and expresses certainty.

Does it express uncertainty, and does anything happen when it is high?

- [ ] Yes — low confidence escalates to a human, automatically
- [ ] It reports confidence; nothing is triggered by it
- [ ] It returns an answer with no stated confidence
- [ ] We do not know

What would invalidate its training data?

A model learns the conditions that generated its data. Name the event that would end those conditions — a market shift, a regulation, a reorganisation, a pandemic.

Ontology — what it is able to see

Which entities it recognises, which relations it models, which variables do not exist for it.

Name one variable that matters to this decision and that the system cannot see.

Geopolitical exposure, a relationship, a regulatory change, a supplier's own fragility. If nothing comes to mind in sixty seconds, that is worth noting as an answer.

Who reviews what the system represents, and when?

- [ ] A named owner, on a fixed cadence
- [ ] Only when something goes wrong

- [ ] Nobody — it has the representation it was built with

5 · The verdict

Where is the largest debt?

- [ ] Technical — the artefact itself
- [ ] Philosophical — unarticulated purpose, knowledge or representation
- [ ] Adoption — it works and people do not use it
- [ ] Capability — we would not be able to improve it after buying it
- [ ] None that we can see

The decision

- [ ] Proceed
- [ ] Proceed once named conditions are met — write the first of them as the step below
- [ ] Pause — the debt has to be paid first
- [ ] Do not proceed

The first step, and who owns it

One line

After the session

Sections 6 and 7 take the appraisal from one tool to the organisation. They are the module's follow-up assignment, and they are better done with a week of thought than with ten minutes of room time.

6 · The reference

Somebody already does this well. The question is not what their number is, but what routine produces it — and what you would have to accumulate before that routine would work here.

Who, or what standard?

The most useful reference is often outside your industry. Xerox fixed its warehouse by studying a catalogue retailer, not a competitor.

An organisation, or a published standard

What does the reference do that you do not?

The practice, not the indicator. "They are 30% faster" is a number; "they review the representation quarterly with a named owner" is a practice.

Would you be copying an asset, a process, or a rule-in-use?

- [ ] A — an asset we would buy
- [ ] P — a process we would document and copy
- [ ] C — a way of working that lives in people's habits, which we would have to build

What would have to be true here for that practice to work?

What would our people need to already know, or already be able to do, for this to take here? Practices fail to transfer for reasons of knowledge far more often than of motivation — including between two units of the same company.

How long did the reference take to build it?

- [ ] Months
- [ ] A few years
- [ ] A decade or more
- [ ] Unknown — nobody here knows

Adopt, adapt or discard

- [ ] Adopt — conditions are already in place
- [ ] Adapt — translate it to our context
- [ ] Discard — it depends on conditions we do not have and cannot reasonably accumulate

7 · Capability gaps and the six-month roadmap

The appraisal above concerns one tool. A roadmap concerns the organisation. Name the three capabilities whose absence most constrains performance — capabilities, not products: "a single authoritative home for schedule dates" rather than "a new PPM system" — then place each one on

the two axes that decide sequence. **Strategic value**: how much it moves something customers choose you for, or removes a constraint that keeps recurring. **Feasibility**: whether the conditions to build it exist today — the people, the data, and the authority to change the process.

Gap 1

The capability that is missing, in one line

Gap 1 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

Gap 2

The capability that is missing, in one line

Gap 2 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

Gap 3

The capability that is missing, in one line

Gap 3 — strategic value, feasibility, and when

- [ ] High
- [ ] Medium
- [ ] Low

What has to be true by the end of month two?

One sentence, measurable. A roadmap whose first checkpoint cannot be stated is a list of intentions with dates attached.

THREE

Reading the result

The instrument has no score, deliberately. It produces three things worth looking at.

- **Which term you could defend.** Most honest appraisals land on A, or on A and P. That is a correctly identified purchase, not a failure. The failure is claiming C and being unable to say, in one sentence, what the organisation will be able to do afterwards without the system.
- **Where the chain first stops being a clear yes.** That link is the finding. If it is condition 2 or 3, the problem is the organisation's and no vendor can fix it. If the answer is "not measured", the next step is to measure, not to spend — an unobserved implementation is a far more tractable problem than a failing one.
- **Where the tool contains AI, how many pillars nobody wrote down.** Not how good the system is — whether the assumption was articulated. Teleology is cheap to write down before deployment and expensive afterwards, once the system has been acting on the implicit hierarchy for two years.

FOUR

Sources

- Schrage, M., & Kiron, D. (2025). Philosophy eats AI. *MIT Sloan Management Review*. The three pillars — teleology, epistemology and ontology — and the argument that the binding constraint on value from AI is philosophical rather than technical, are theirs.
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- Van Mieghem, J. A., & Allon, G. (2015). *Operations Strategy: Principles and Practice* (2nd ed.). Dynamic Ideas.
- Lima, T. F. (2026). *Excelência operacional*, ch. 1, for the identity V = C × (A + P).

The mapping between the three pillars and the operational arguments of the session — teleology to order winners, epistemology to signalling at the source, ontology to causal ambiguity — is this course's, not the original authors'.

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